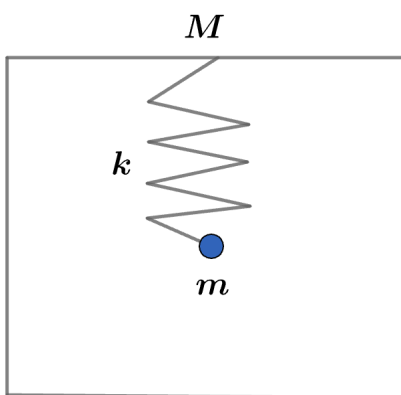


2017 F=ma Exam: Problem 2

Kevin S. Huang

When the box is held fixed, we have a standard mass-spring system which has angular frequency

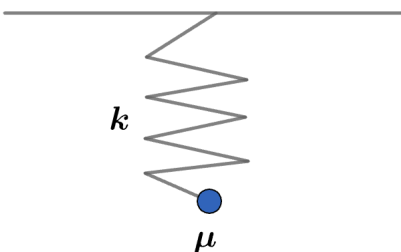
$$\omega = \sqrt{\frac{k}{m}}$$



Once the box is dropped, we have a coupled oscillator of two masses m and M connected by a spring of constant k . Recall the reduced mass is given by

$$\frac{1}{\mu} = \frac{1}{m} + \frac{1}{M}$$
$$\mu = \frac{mM}{m+M}$$

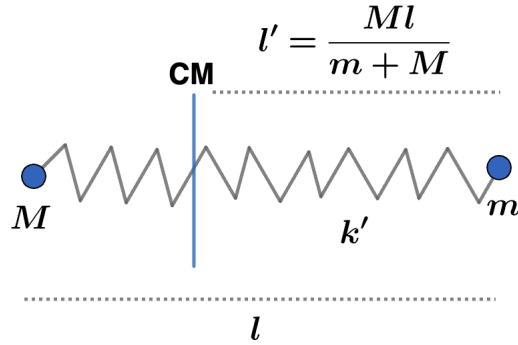
The reduced mass tells us the mass of the particle in the equivalent one-body problem to the original two-body problem. This single particle should be under the influence of the same force as that between the two bodies. In our case, we have mass μ attached to a spring with constant k .



The angular frequency is then

$$\omega' = \sqrt{\frac{k}{\mu}} > \omega$$

since $\mu < m, M$. Alternatively, we can also obtain ω' directly, by noting that both masses oscillate around their CM.



Mass m is effectively connected to a shortened spring of length

$$l' = \frac{Ml}{m+M}$$

Since $k \propto 1/l$, the effective spring constant k' is

$$k' = \frac{kl}{l'} = \frac{(m+M)k}{M}$$

Then

$$\omega' = \sqrt{\frac{k'}{m}} = \sqrt{\frac{(m+M)k}{mM}} = \sqrt{\frac{k}{\mu}}$$

as required. Thus, the answer is B.